POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2017
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define signal processing? Compare between energy signal and power signal. 7

   b) State and prove the time-shifting property of DTFS. 8

2. a) If the impulse response to a LTI system is given as 7

   $h[n] = \{1\ 2\ 1\}$

   and the output to that system is

   $y[n] = \{1\ 3\ 4\ 3\ 1\}$

   Determine the input signal of that system.

   b) Define causal and non-causal system. Prove that a LTI system is stable if and only if its impulse response is absolutely summable. 8

3. a) Define z – transform and Region of Convergence. Find z – transform of following signal $x(n) = a^n \sin w_0 n\, u(n)$. Also determine the ROC of the signal. 7

   b) Compute the eight-point DFT of the sequence 8

   $$x[n] = \begin{cases} 1, & 0 \le n \le 7 \\ 0, & otherwise \end{cases}$$

   by using the decimation in frequency FFT algorithm.

4. a) Given a three stage lattice filter with coefficients $k_1 = 1/2$, $k_2 = 1/4$ and $k_3 = 1/4$. Determine the FIR filter coefficients for the direct form structure. 7

   b) Obtain the direct form realizations for the system described by the following equation: 8

   $2y(n) + y(n-1) - 4y(n-5) = x(n) + 3x(n-4)$

5. a) Why ideal lowpass filter cannot be realized in practice? Explain how practical lowpass filter are realized in practice and also explain its 7

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effect.

b) Design a linear FIR filter using Kaiser window to meet the following specifications:

$$0.98 \leq |H(e^{j\omega})| \leq 1.02, \quad for\ 0 \leq |\omega| \leq 0.19\pi$$

$$|H(e^{j\omega})| \leq 0.02, \quad for\ 0.21\pi \leq |\omega| \leq \pi$$

6. a) Use the bilinear transformation to convert the analog filter with system function

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter. Select T = 0.1 and compare the location of the zeros in H(z) with the locations of the zeros obtained by applying the impulse invariance method in the conversion of H(s).

b) Design a digital lowpass Butterworth filter using Bilinear Transformation method to meet the following specifications:

Passband attenuation ≤ 1 dB
Stopband attenuation ≥ 30 dB
Passband edge = 1.5 KHz
Stopband edge = 4 KHz
Sampling frequency = 10 KHz

7. Write short notes on: **(Any two)**

a) Digital Signal Processing versus Analog Signal Processing

b) Time shifting and Frequency shifting properties of Fourier Transform

c) Gibb's phenomenon

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